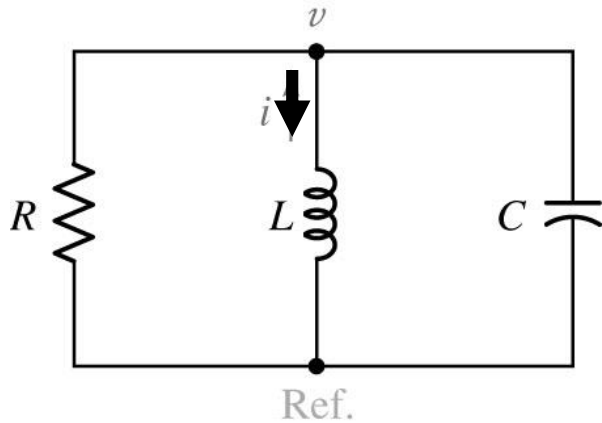


The RLC Circuit

Differential Equation for Parallel RLC Circuit



Nodal equation:

$$\frac{v}{R} + \frac{1}{L} \int_{t_0}^t v dt' + i(t_0) + C \frac{dv}{dt} = 0$$

$$I(0^+) = I_0$$

$$V(0^+) = V_0$$

$$C \frac{d^2 v}{dt^2} + \frac{1}{R} \frac{dv}{dt} + \frac{1}{L} v = 0$$

Solution $v(t)$ – desired natural response

$$v(t) = Ae^{st}$$

$$Cs^2 + \frac{1}{R}s + \frac{1}{L} = 0$$

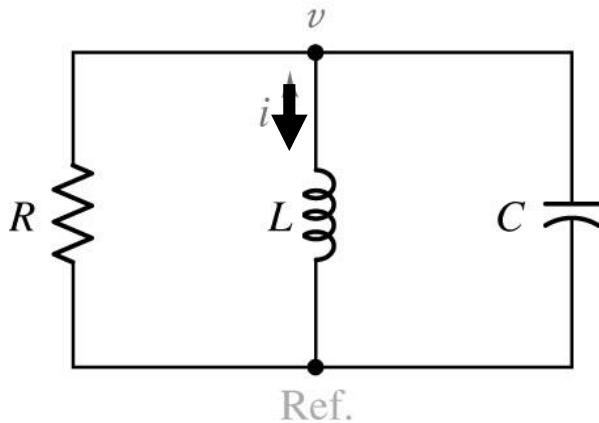
$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Definitions

$$C \frac{d^2 v}{dt^2} + \frac{1}{R} \frac{dv}{dt} + \frac{1}{L} v = 0$$

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$



$$\alpha = \frac{1}{2RC}$$

**neper frequency or
differential damping coefficient**

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

resonant frequency

s_1, s_2 – complex frequencies

$\alpha > \omega_0$, s_1, s_2 – real numbers – overdamped response

$\alpha < \omega_0$, s_1, s_2 – imaginary numbers – underdamped response

$\alpha = \omega_0$ – critically damped response

Damping

- **Undamped:** The system oscillates at its natural resonant frequency (ω_0).
- **Overdamped:** The system returns (exponentially decays) to equilibrium without oscillating.
- **Critically damped:** The system returns to equilibrium *as quickly as possible* without oscillating.
- **Underdamped:** The system oscillates (at reduced frequency compared to the undamped case) with the amplitude gradually decreasing to zero.

Example

Consider parallel RLC circuit having an inductance 10 mH and a capacitance of 100 μ V. Determine the resistor values that would lead to overdamped and underdamped responses

Resonant frequency $\omega_0 = \sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{(10 \cdot 10^{-3})(100 \cdot 10^{-6})}} = 10^3 \text{ rad/s}$

For **overdamped response** $\alpha > \omega_0$;

for **underdamped response** $\alpha < \omega_0$

$$\frac{1}{2RC} > 10^3$$

$$R < 5 \Omega$$

$$\frac{1}{2RC} < 10^3$$

$$R > 5 \Omega$$

Practice

A parallel RLC circuit contains a $100\ \Omega$ resistor and has parameter values $\alpha=1000\ \text{s}^{-1}$ and $\omega_0=800\ \text{rad/s}$. Find:
 C ; L ; s_1 ; s_2

Overdamped Parallel RLC Circuit

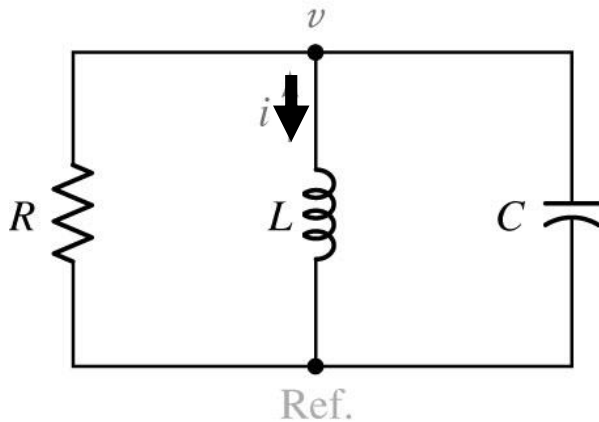
$\alpha > \omega_0$, s_1, s_2 – real numbers – **overdamped response** $LC > 4R^2C^2$

$$\sqrt{\alpha^2 - \omega_0^2} < \alpha$$

$$(-\alpha - \sqrt{\alpha^2 - \omega_0^2}) < (-\alpha + \sqrt{\alpha^2 - \omega_0^2}) < 0$$

s_1, s_2 – are real **negative** numbers

$$|s_2| > |s_1|$$



s_2 has more rapid speed of decay

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$v(t) \rightarrow A_1 e^{s_1 t} \rightarrow 0 \text{ as } t \rightarrow \infty$$

$$v(0) = A_1 + A_2 = \dots$$

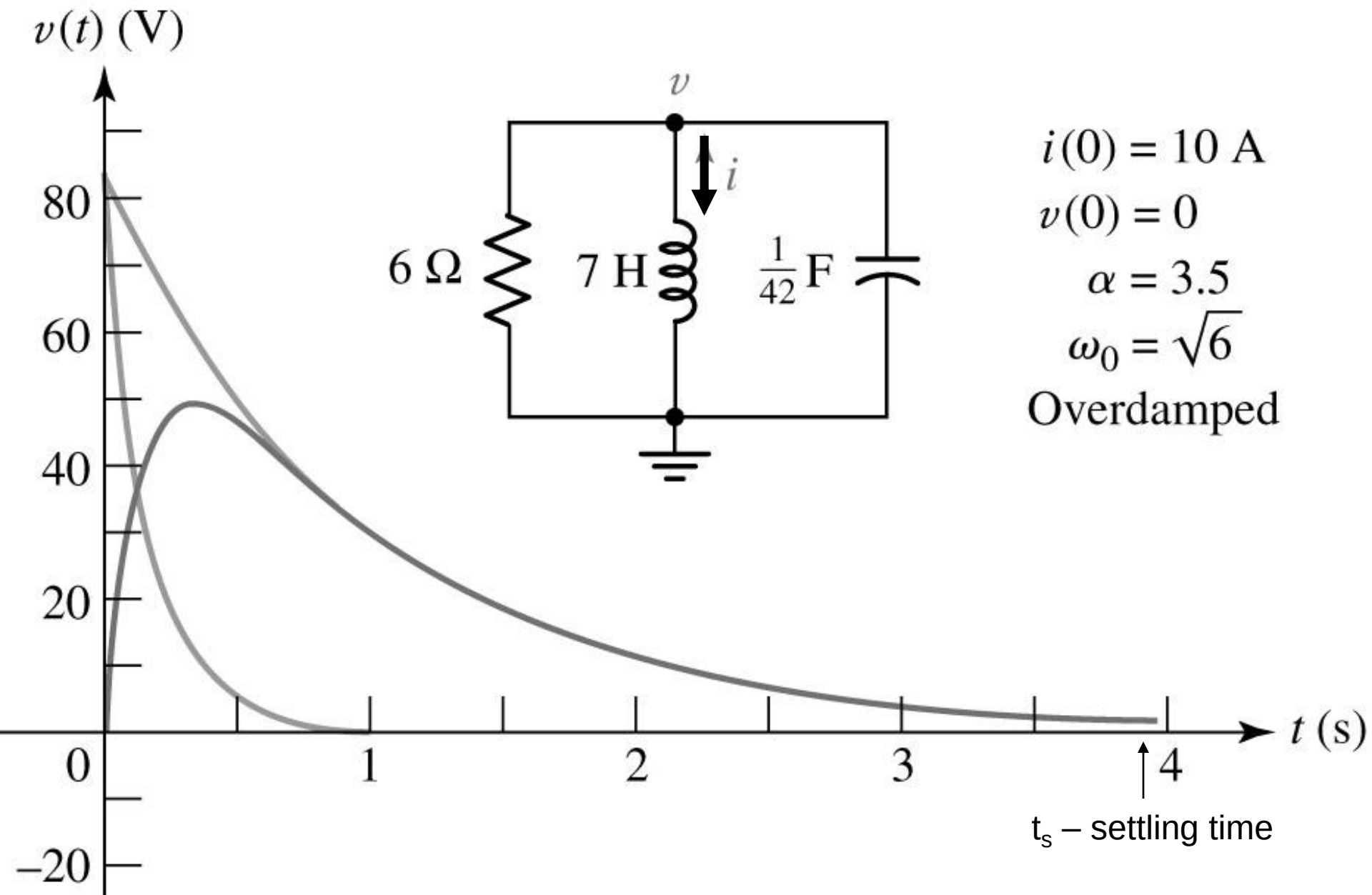
$$\left. \frac{dv}{dt} \right|_{t=0} = \dots$$

$$i_C = C \frac{dv}{dt}$$

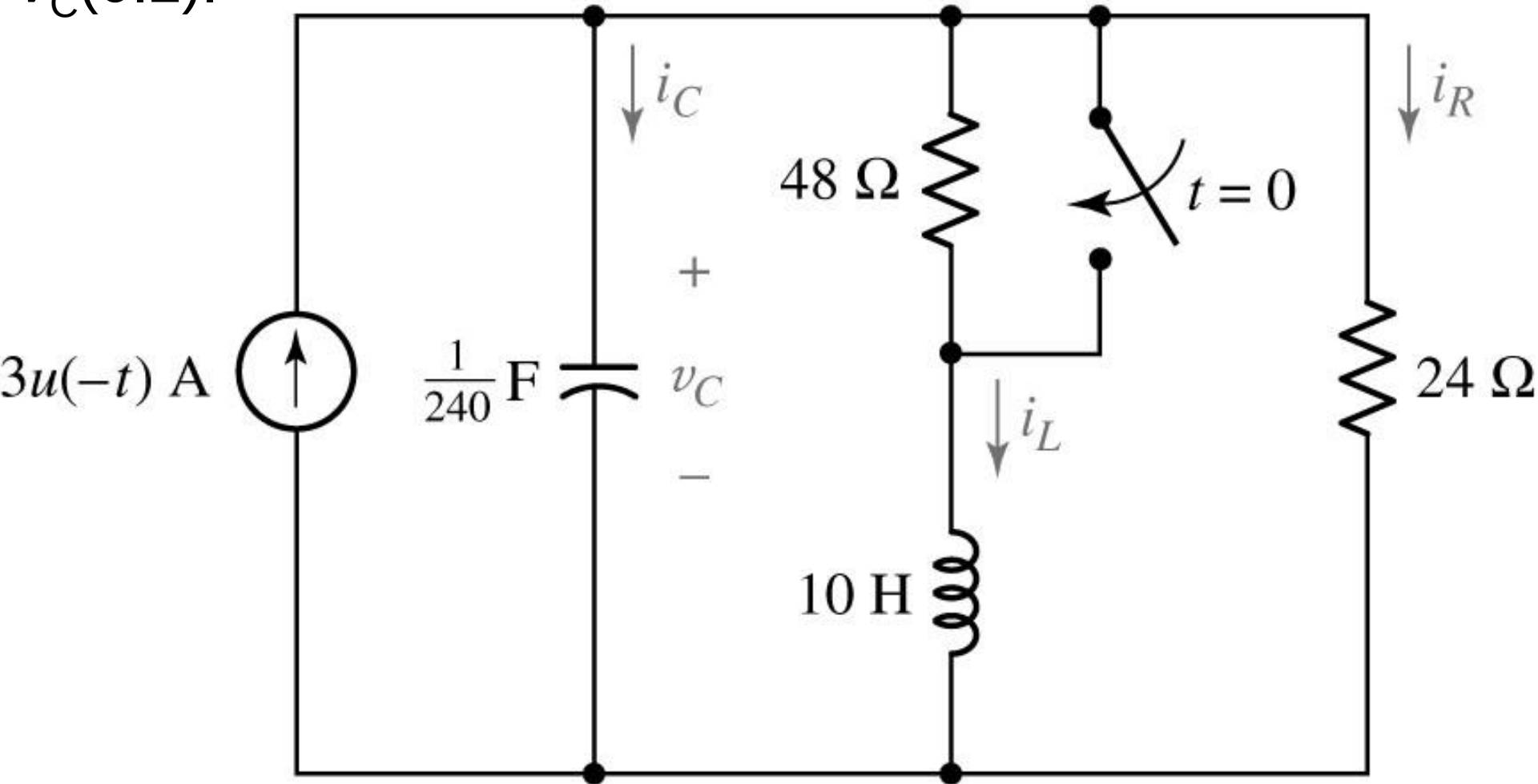
$$i_C + i_L + i_R = 0$$

$$\left. \frac{dv}{dt} \right|_{t=0} = \frac{i_C}{C} = \dots$$

Overdamped Parallel RLC Circuit – Graphically

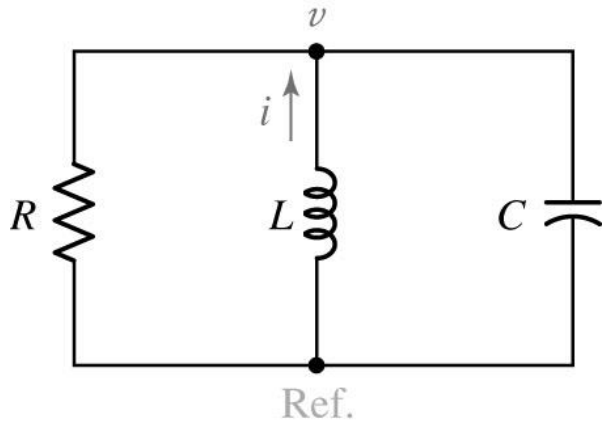


After being open for a long time, the switch closes at $t=0$. Find (a) $i_L(0^-)$; (b) $v_C(0^-)$; (c) $i_R(0^+)$; (d) $i_C(0^+)$; (e) $v_C(0.2)$.



1A; 48 V; 2A; -3A; -17.54V

Critically Damped Parallel RLC Circuit



$$\begin{cases} \alpha = \omega_0 \\ LC = 4R^2C^2 \\ L = 4R^2C \end{cases}$$

$$\alpha = \frac{1}{2RC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$C \frac{d^2v}{dt^2} + \frac{1}{R} \frac{dv}{dt} + \frac{1}{L} v = 0$$

$$\frac{d^2v}{dt^2} + 2\alpha \frac{dv}{dt} + \alpha^2 v = 0$$

$$v(t) = e^{-\alpha t} (A_1 t + A_2)$$

$$v(0) = A_2$$

$$\text{If } v(0) = 0 \quad A_2 = 0$$

$$\frac{dv}{dt} = -\alpha e^{-\alpha t} (A_1 t + A_2) + A_1 e^{-\alpha t}$$

$$\text{If } v(0) = 0 \quad \left. \frac{dv}{dt} \right|_{t=0} = A_1$$

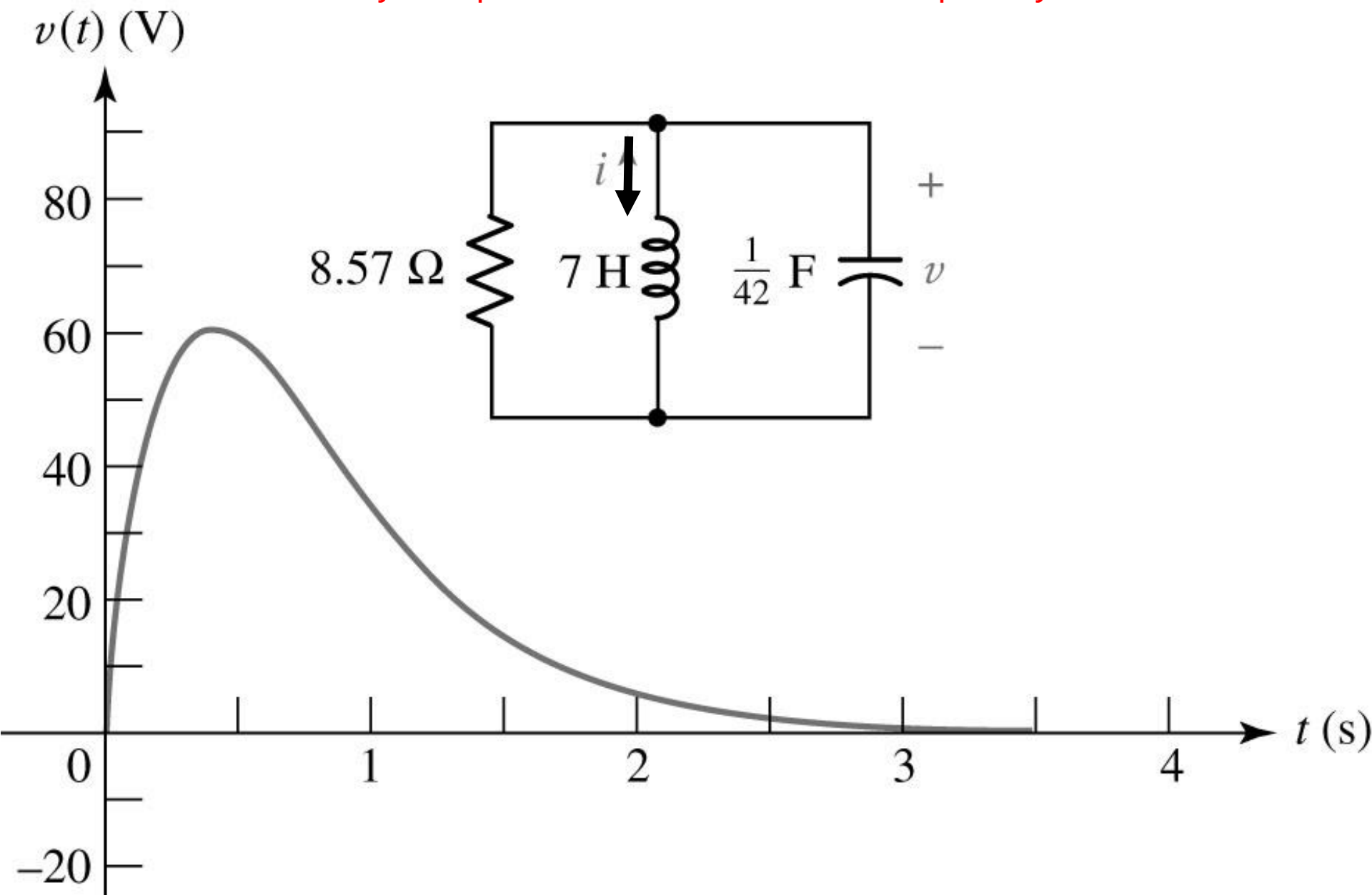
$$\left. \frac{dv}{dt} \right|_{t=0} = -\alpha A_2 + A_1$$

$$i_C = C \frac{dv}{dt}$$

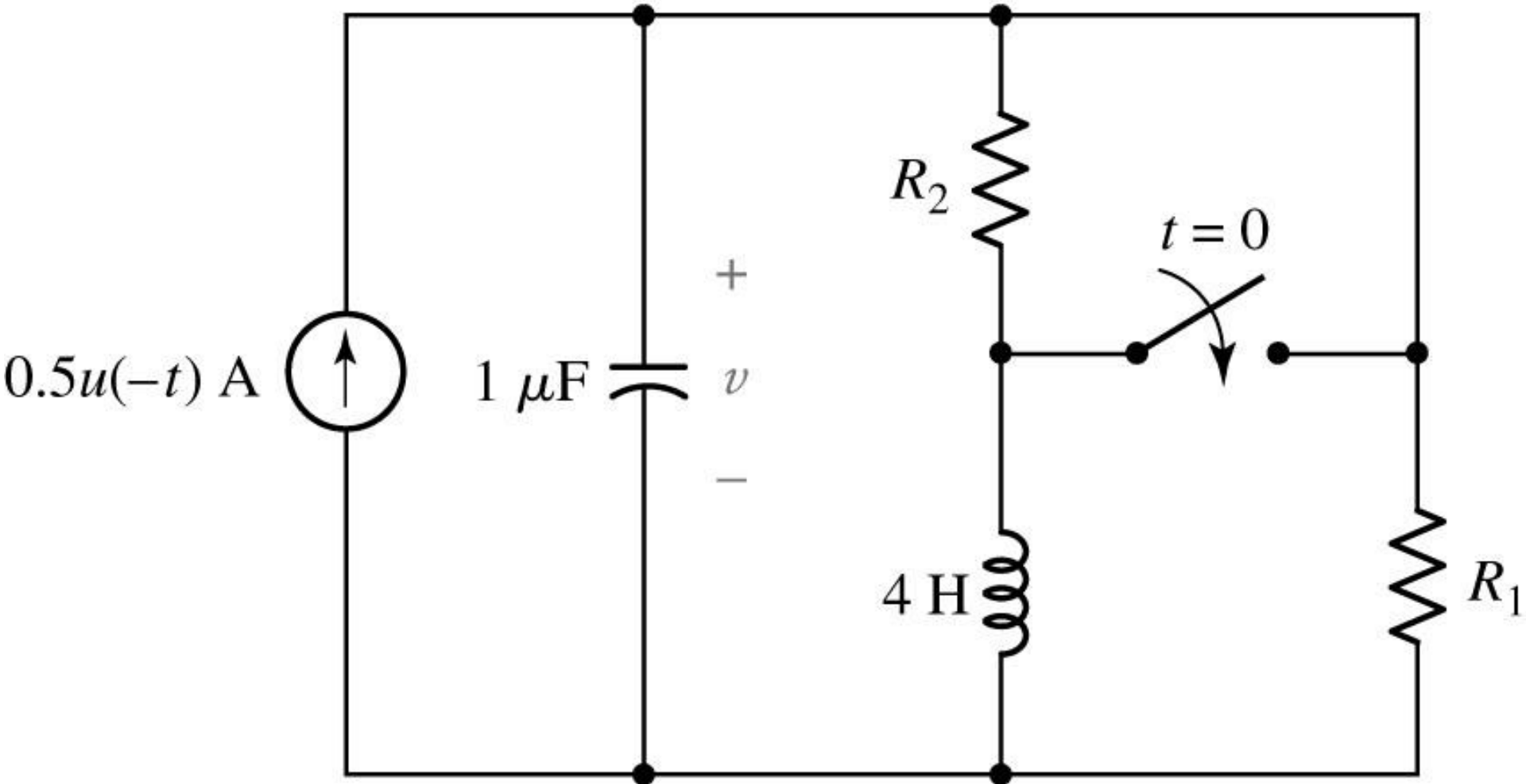
$$i_C + i_L + i_R = 0$$

$$\left. \frac{dv}{dt} \right|_{t=0} = \frac{i_C}{C} = \dots$$

Critically Damped Parallel RLC Circuit – Graphically

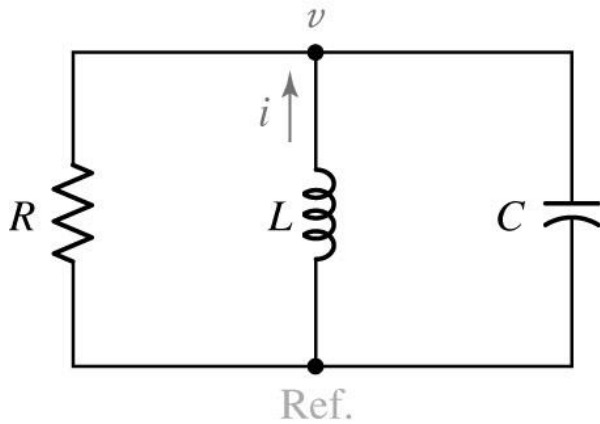


(a) Choose R_1 in the circuit so that the response after $t=0$ will be critically damped. (b) Now select R_2 to obtain $v(0)=100\text{V}$. (c) Find $v(t)$ at $t=1\text{ ms}$.



$1\text{ k}\Omega$; $250\text{ }\Omega$; -212 V

Underdamped Parallel RLC Circuit



$$\alpha < \omega_0$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$C \frac{d^2 v}{dt^2} + \frac{1}{R} \frac{dv}{dt} + \frac{1}{L} v = 0$$

$$\alpha = \frac{1}{2RC}$$

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$\sqrt{\alpha^2 - \omega_0^2} = \sqrt{-1} \sqrt{\omega_0^2 - \alpha^2} = j \sqrt{\omega_0^2 - \alpha^2}$$

Natural resonant frequency: $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

$$v(t) = e^{-\alpha t} (A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t})$$

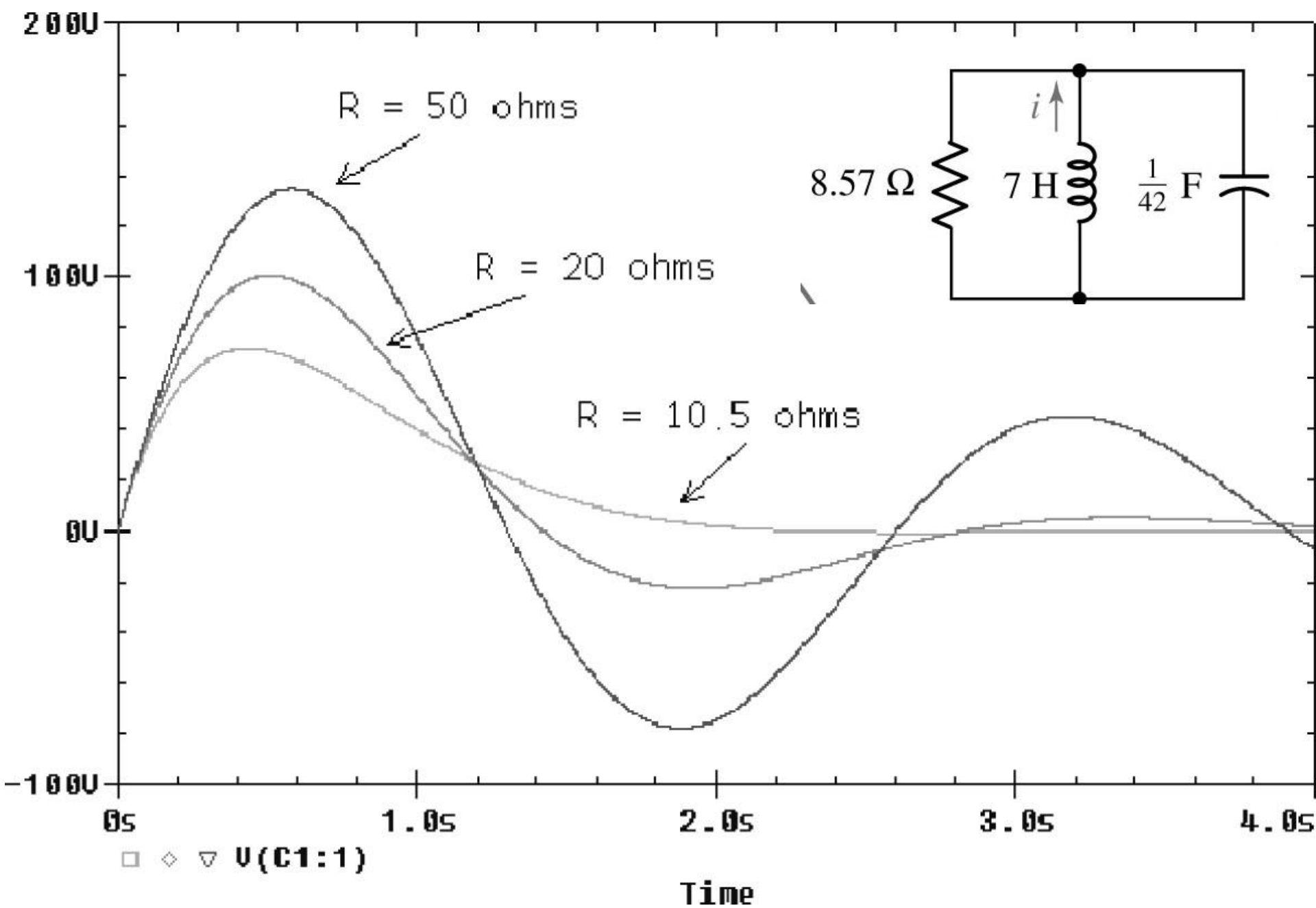
$$v(t) = e^{-\alpha t} \left((A_1 - A_2) \frac{e^{j\omega_d t} + e^{-j\omega_d t}}{2} + j(A_1 + A_2) \frac{e^{j\omega_d t} - e^{-j\omega_d t}}{2j} \right)$$

$$v(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

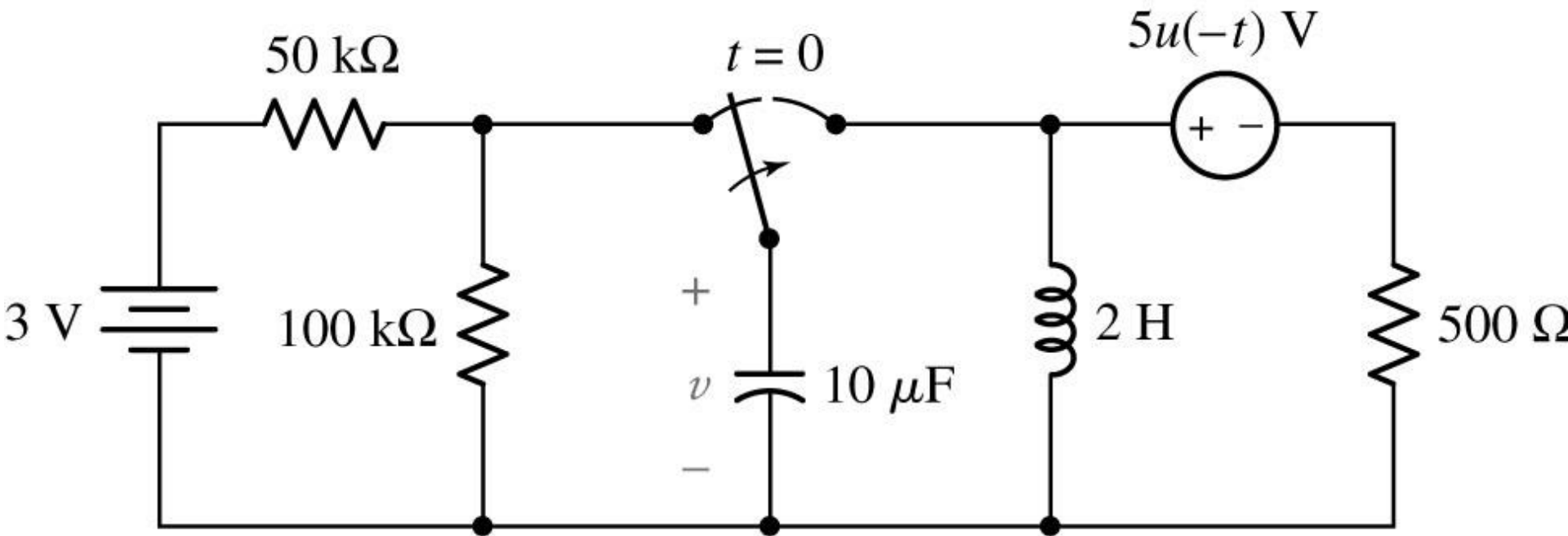
$$v(0) = \dots$$

$$\left. \frac{dv}{dt} \right|_{t=0} = \dots$$

Underdamped Parallel RLC Circuit – Graphical



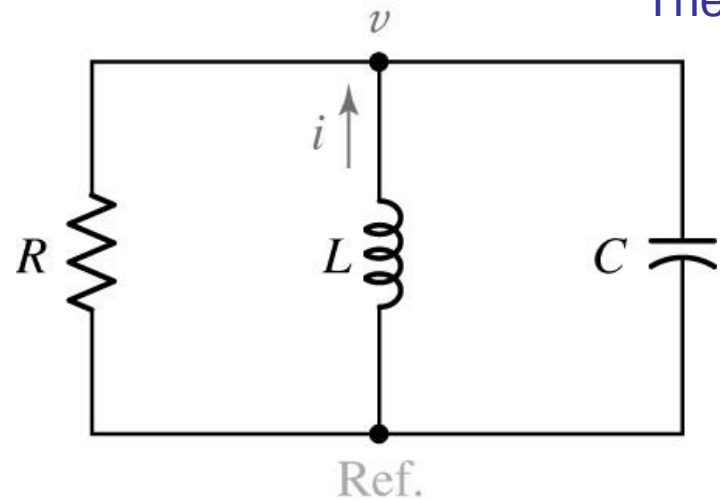
After being in the left position for a long time, the switch moves to the right at $t=0$. Find (a) dv/dt at $t=0^+$; (b) v at $t=1\text{ ms}$; (c) t_0 the first value of t greater than 0 at which $v=0$.



-1400 V/s; 0.695 V; 1.609 ms

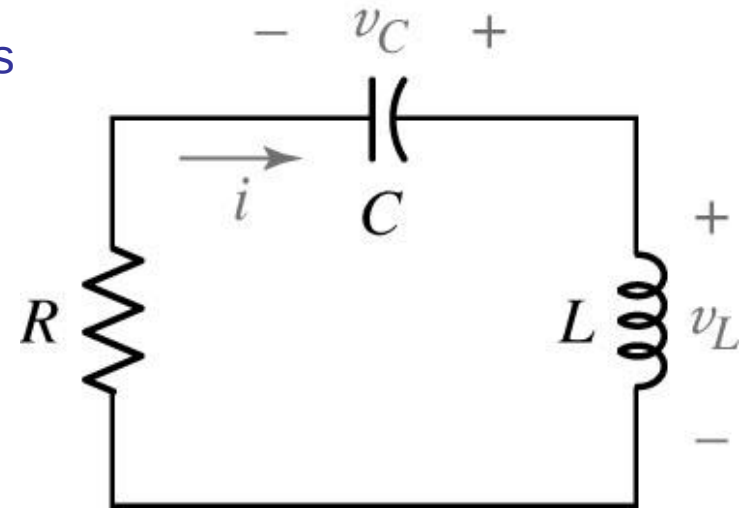
Source-free Series RLC Circuit

They are dual circuits



$$C \frac{d^2 v}{dt^2} + \frac{1}{R} \frac{dv}{dt} + \frac{1}{L} v = 0$$

$$\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$



$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = 0$$

$$\alpha = \frac{R}{2L} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Overdamped

$$\alpha > \omega_0$$

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Critically damped

$$\alpha = \omega_0$$

$$i(t) = e^{-\alpha t} (A_1 t + A_2)$$

Underdamped

$$\alpha < \omega_0$$

$$i(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t) \quad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

Complete response of RLC Circuit

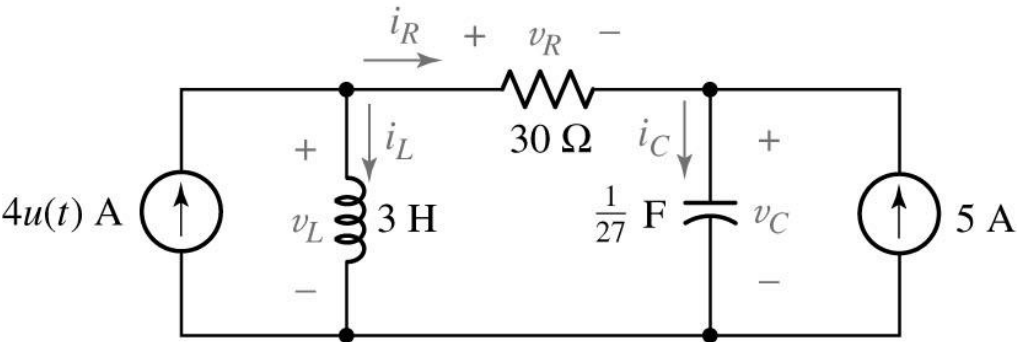
As in the case of RL and RC, complete response is

$$v(t) = v_f + v_n$$

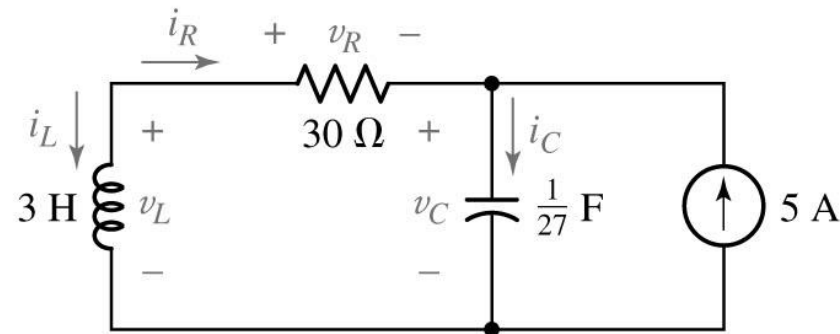
$$v_n(t) = Ae^{s_1 t} + Be^{s_2 t} \qquad s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$v(0^+) = v_f + A + B \qquad \frac{dv}{dt} = 0 + As_1 e^{s_1 t} + Bs_2 e^{s_2 t}$$

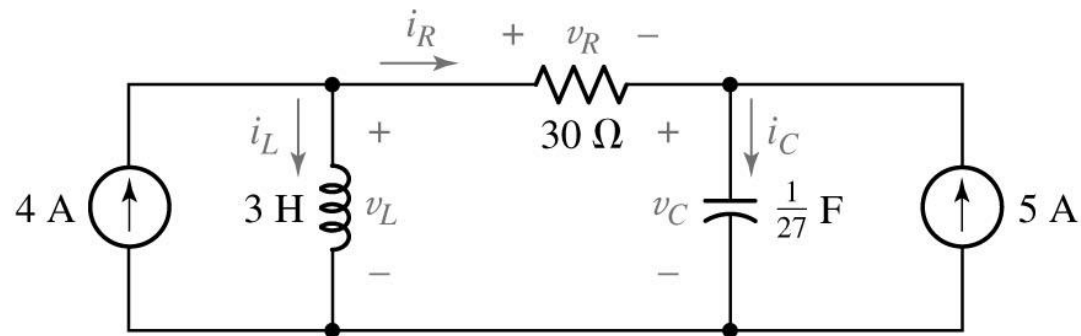
There are three passive elements in the circuit and a voltage and a current are defined for each. Find the values of these six quantities at both $t=0^-$ and $t=0^+$.



(a)



(b)



(c)

Let $i_s = 10u(-t) - 20u(t)$ A. Find (a) $i_L(0^-)$; (b) $v_C(0^+)$; (c) $v_R(0^+)$; (d) $i_L(\infty)$; (e) $i_L(0.1 \text{ ms})$

